

Replication Instructions

This document provides the steps required to reproduce the experimental results reported in the coursework report for the proposed evolutionary search tool and the Random Search baseline.

Step 1: Install dependencies

Ensure Python 3.x is installed along with the required libraries listed in requirements.pdf:

- scipy
- pandas
- numpy
- matplotlib

Step 2: Prepare the project structure

Clone or download the GitHub repository and ensure the following folders exist in the root directory:

- datasets/
- search_results/
- visualization_results/

Step 3: Run the comparison experiments

`python comparison.py`

This is the main script used to reproduce the results reported in the coursework report. It performs 30 repetitions for both the proposed tool and the Random Search baseline using:

- measurement budget = 100
- initial population percentage = 20%
- fixed random seeds for reproducibility

The script generates:

- raw 30 run result files
- final_results.csv containing the summary table used in the report, which also includes the statistical test results

Optional step : Run the proposed tool separately

`python tool.py`

This runs the evolutionary search tool independently on all datasets and prints the best configuration and best performance found for each system.

Step 4: Generate boxplots

python boxplot.py

This generates the boxplot figures used in the report and stores them inside the visualization_results/ folder.

Step 5: Verify results

The generated final_results.csv should match the values presented in Table 2 of the report, including:

- average performance values
- improvement percentages
- Mann-Whitney U test p-values

The generated boxplots should also match the figures presented in the Experiments section of the report, or the visualization_results folder in the repository, for comparison of all the boxplots.

Notes

- Lower performance values indicate better configurations for all datasets
- All experiments are performed as minimisation problems
- Random seeds are fixed using random.seed(42) and numpy.random.seed(42) to ensure reproducibility
- Small numerical differences may occur depending on Python version or library versions

The table used in the report is given below for reference

System	Baseline Avg	Tool Avg	Tool Std	Imp. %	P-Value
7z	4708.21	4638.55	819.13	1.48%	0.0276*
Apache	31.05	30.90	0.16	0.51%	0.0039*
LLVM	58634.27	55785.13	1448.53	4.86%	< 0.001*
Postgres	46043.11	46017.75	65.72	0.06%	0.1768
Brotli	1.48	1.48	0.02	0.33%	0.9184
Spear	0.0011	0.0011	0.0009	3.03%	0.6139
Storm	0.00001	0.0000	0.00	67.96%	0.1469
x264	22.65	21.85	0.41	3.54%	< 0.001*